



Early-Stage Diabetes Risk Prediction

Using Random Forest & XGBoost Classifiers

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Why Early Detection Matters

- *Diabetes is a major health concern affecting daily life and quality of care*
- *Many people don't recognize symptoms early or ignore them until serious*
- *Early identification enables timely medical intervention*

The Opportunity

- *Healthcare data is increasingly available*
- *Machine Learning can identify patterns in patient symptoms and factors*
- *Structured prediction can support early screening decisions*



*Goal: Predict possible diabetes risk from patient information using ML
– as an early **screening support tool**, not a medical diagnosis.*

Project Objectives

1

Predict

Develop an ML system for early-stage diabetes risk prediction

2

Preprocess

Prepare patient data for model training – handle categorical values and check data quality

3

Compare

Train and compare Random Forest and XGBoost classifiers side by side

4

Evaluate

Measure model performance using accuracy, precision, recall, F1-score, and confusion matrix

5

Identify

Recommend model performs better on the diabetes dataset

Understanding the Data

520

Patient Records

UCI Early Stage Diabetes Risk Prediction Dataset

16

Input Features

Patient characteristics and clinical symptoms

2

Target Classes

Positive (at risk) vs. Negative (not at risk)

Patient Features Include

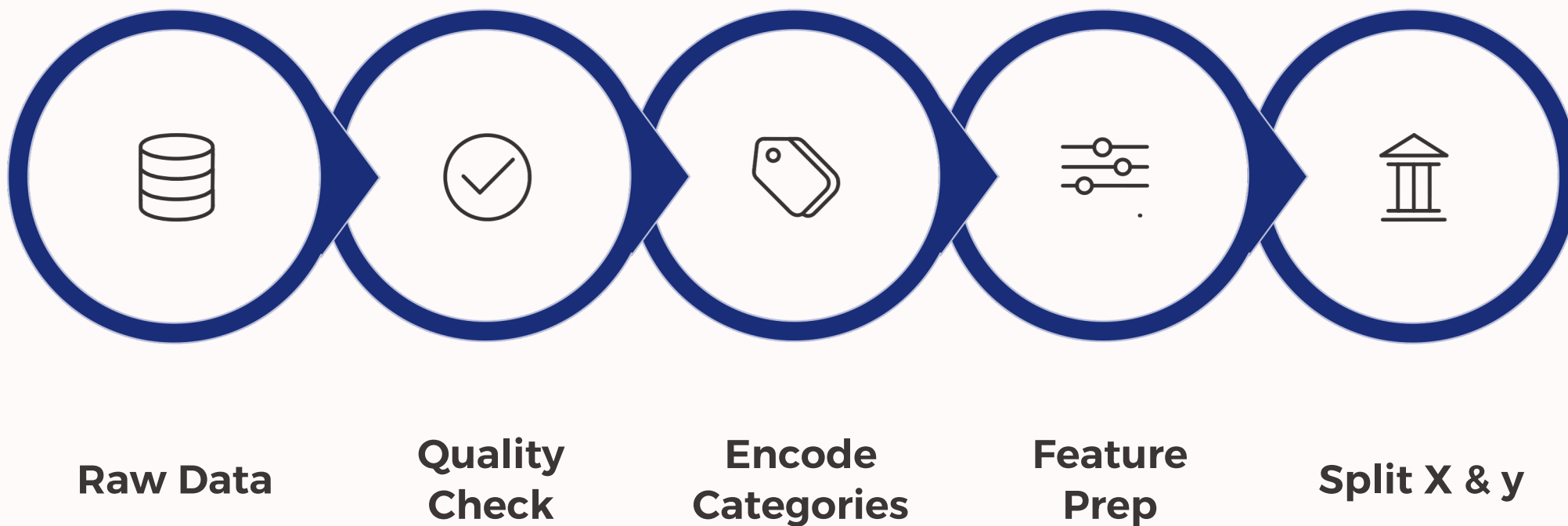
- *Age & Gender*
- *Polyuria & Polydipsia*
- *Sudden Weight Loss*
- *Weakness & Obesity*

Additional Symptoms

- *Visual Blurring*
- *Itching & Irritability*
- *Delayed Healing*
- *Partial Paresis & Alopecia*

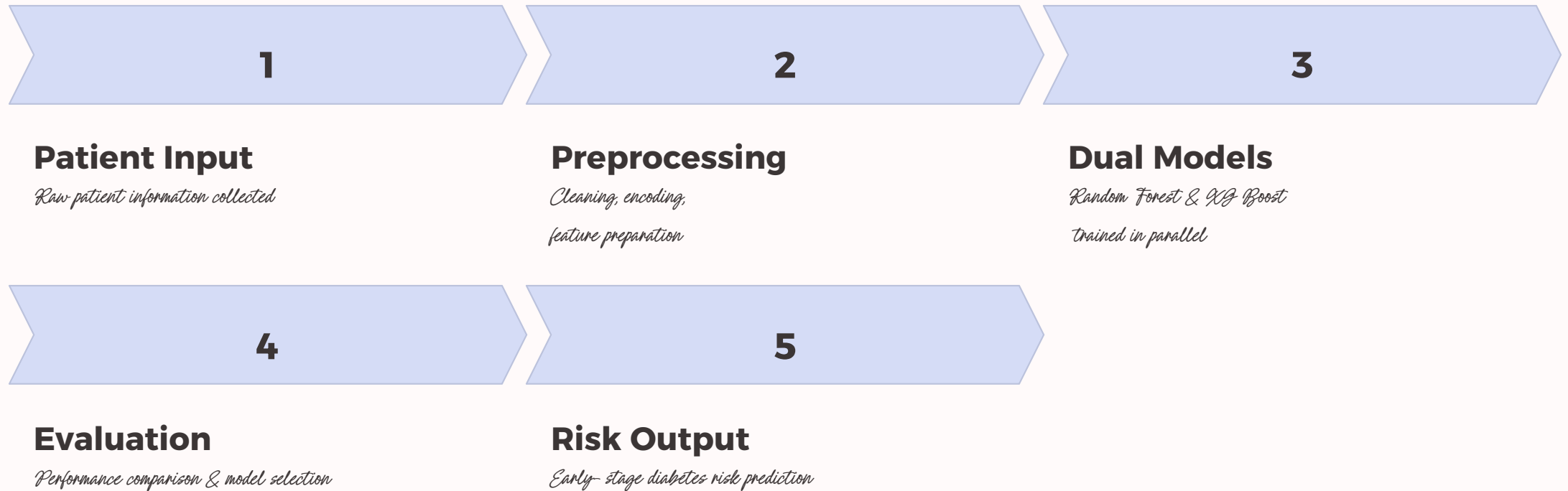
These patient characteristics and symptoms become inputs to the ML models for risk prediction.

Data Preprocessing Pipeline



Each preprocessing step ensures the patient data is clean, consistent, and correctly structured before being fed into either classifier – a critical foundation for reliable model training.

Proposed ML Pipeline Architecture





Model 1 – Random Forest

Core Concept: An ensemble approach that combines predictions from multiple decision trees to produce a robust final classification.

Key Mechanisms

Bagging

Each tree is trained on a random subset of the data

Feature Subsampling

Each split considers a random subset of features

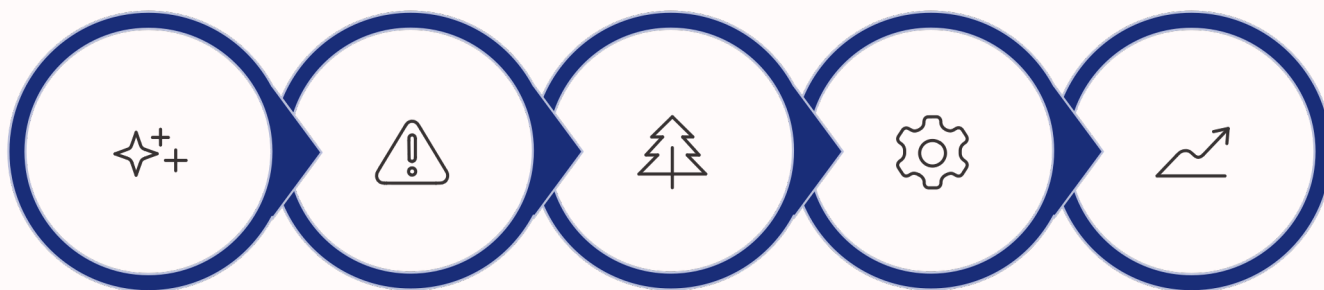
Variance Reduction

Combining multiple trees reduces overfitting and improves generalization

Model 2 – XGBoost

Core Concept: A boosting approach that builds models sequentially – each tree focuses on correcting the errors made by the previous one.

Sequential Learning Process



**Initial
Prediction**

**Identify
Errors**

**Correct
with Tree 2**

**Refine with
Tree 3**

**Improved
Model**

Key Advantage

Focused Learning

*Sequential learning directs
computational effort toward difficult, misclassified
cases*

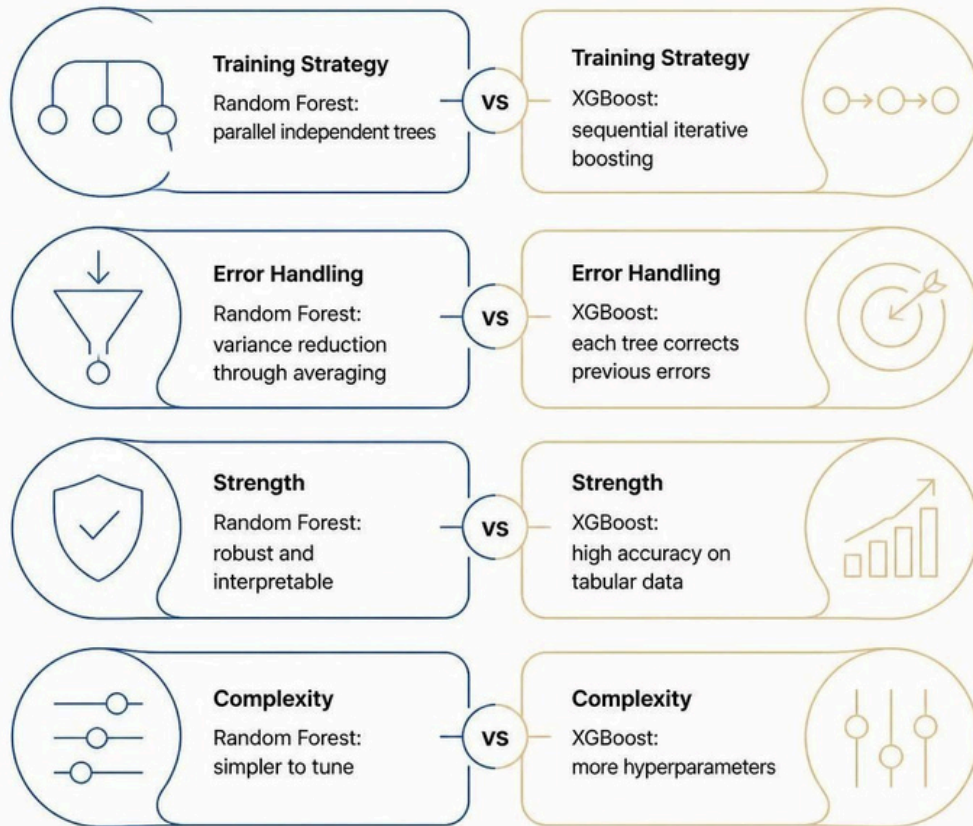
Tabular Data Strength

*XGBoost consistently achieves
strong performance on structured
clinical datasets*

Iterative Refinement

*Each additional tree progressively
reduces prediction error*

Why Compare Two Models?



Fair Comparison Protocol

- Both models trained on the *same preprocessed dataset*
- Evaluated using the *same metrics*
- Tested under *identical conditions*

A controlled comparison ensures the results reflect genuine differences in model capability – not differences in data preparation or evaluation approach.

 *The better-performing model will be selected as the recommended screening classifier.*

Model Evaluation Metrics



Accuracy

Overall proportion of correct predictions across both classes



Precision

Of predicted positive cases, how many are actually positive – reliability of positive predictions



Recall

Of all actual positive cases, how many are detected – sensitivity to true positives



F1-Score

Harmonic mean of precision and recall – balances both metrics for a single performance figure



Confusion Matrix

Distribution of true positives, true negatives, false positives, and false negatives

Implementation Technology Stack



Python 3.x

*Primary
programming
language*



Jupyter / Colab

*Development &
experimentation
environment*



Pandas & NumPy

*Data manipulation
and numerical
computations*



Matplotlib & Seaborn

*Data visualization
and exploratory
analysis*



Scikit-learn

*Random Forest,
preprocessing, and
evaluation metrics*



XGBoost

*XGBoost classifier
implementation*



System Requirements

- *Intel Core i3+ processor*
- *4 GB RAM minimum*
- *2 GB available storage*
- *Standard academic or personal computer – sufficient for this dataset size*



*All tools are open-source and
freely accessible for academic
use.*

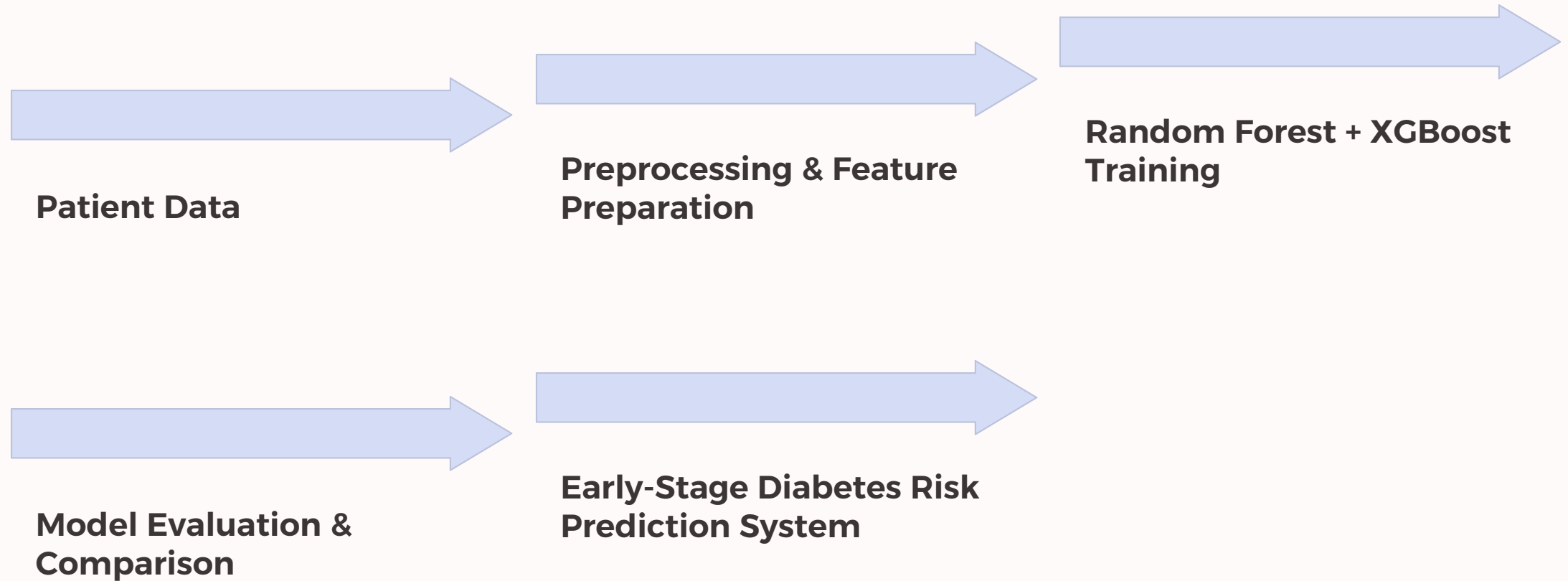
Expected Outcomes & Limitations

- *Develop a complete early-stage diabetes risk prediction workflow*
- *Train Random Forest and XGBoost models on the UCI dataset*
- *Evaluate both models using accuracy, precision, recall, F1-score, and confusion matrix*
- *Compare model performance and identify the better-performing classifier*
- *Demonstrate understanding of ensemble methods and model evaluation*

Important Limitation:

This system is designed as an early screening support tool and is NOT a replacement for professional medical diagnosis. The ML model provides an indication based on available patient information and should always be followed by professional medical evaluation.

Conclusion & Next Steps



Key Insight

Patient symptoms and identifying possible diabetes risk at an early stage.

Future Directions:

- *Implementation and experimentation with actual results*
- *Model comparison using real performance metrics*
- *Further validation on additional datasets*
- *Explainable predictions and practical interface development*



THANK YOU